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LAB 1: REQUIREMENTS ENGINEERING & UML

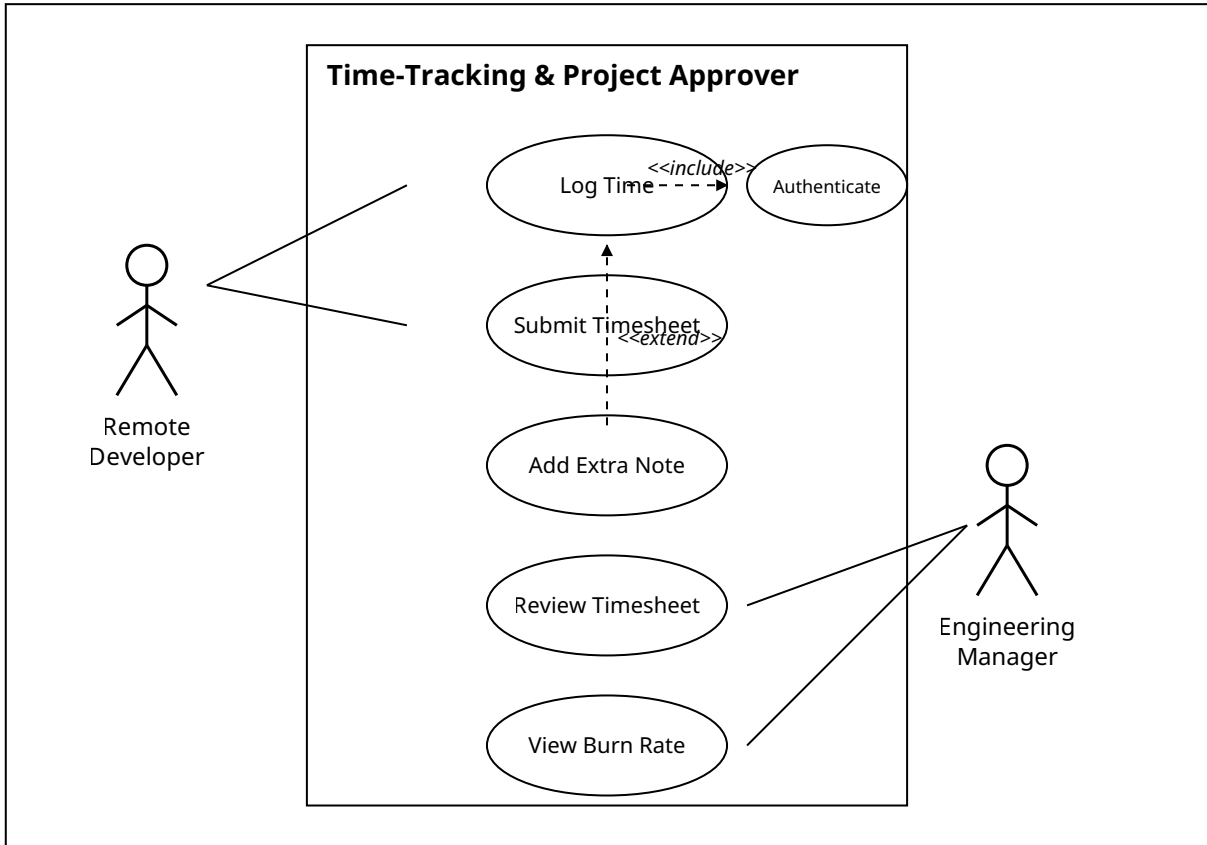
Problem Statement #46: Remote Team Time-Tracking & Project Approver

1. Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow developers to log time against specific project tasks and prevent timesheet submissions exceeding 24 hours in a single calendar day.	High	Pass: Valid timesheet submitted for manager approval. Fail: Timesheet logging >24 hours in one day accepted.	Ensures realistic time logging bounded by physical daily limits.
FR-002	Functional	The system shall allow developers to submit aggregated weekly timesheets to their designated Engineering Manager.	High	Pass: Manager dashboard updates with pending timesheet notification. Fail: Submission fails silently without notifying manager.	Facilitates the core workflow for payroll and managerial sign-off.
FR-003	Functional	The Engineering Manager shall be able to approve or reject submitted timesheets and append feedback comments.	High	Pass: Status changes to Approved/Rejected and developer receives notification. Fail: Status remains pending after action.	Provides necessary managerial oversight and error correction.
FR-004	Functional	The system shall calculate and display real-time project budget burn rates by correlating approved logged hours with developer hourly rates.	Medium	Pass: Dashboard burn chart updates accurately upon timesheet approval. Fail: Burn rate diverges from approved cost.	Essential for monitoring project financial health and constraints.
FR-005	Functional	The system shall automatically fetch and synchronize active tasks assigned to the developer from Jira.	Medium	Pass: Developer sees current active Jira tickets mapped accurately. Fail: Tasks are missing or improperly assigned.	Eliminates manual entry errors and ensures consistency with task trackers.

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NFR-001	Perf. & Security	Timesheet approval status transitions and project budget burn charts must update across dashboards within 500 ms.	High	Pass: Benchmarking tests confirm <500ms latency under simulated peak load.	Ensures a smooth, highly responsive user experience.
NFR-002	Reliability	The platform shall maintain 99.9% uptime during regular business hours to guarantee continuous time logging capabilities.	High	Pass: Monitoring tools report less than 43 minutes of downtime per month.	Critical system availability ensures continuous daily operations.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: UC-001 Log Time against Project Task

Primary Actor: Remote Developer

Preconditions: Developer is securely authenticated into the Time-Tracking platform and mapped to active Jira tickets.

Postconditions: The time entry is saved successfully, queued for manager approval, and the daily logged total is updated.

Main Success Scenario:

1. The Developer navigates to the Time-Tracking interface.
2. The System retrieves and displays active, assigned Jira tickets.
3. The Developer selects a task, enters the hours worked, and submits the entry.
4. The System verifies that the newly submitted time does not cause the daily total to exceed 24 hours.
5. The System records the entry as 'Pending Approval' and updates the dashboard.

Alternate Flow (1a: Daily limit exceeded):

- **4a.** The input causes the daily total for the calendar day to exceed 24 hours.
- **4b.** The System rejects the submission, preventing the save action, and displays a validation error: "Daily maximum of 24 hours exceeded."
- **4c.** The Developer reviews, corrects the entered hours, and re-submits.